



## Article 2: Your Factory Is Running. Your Firewall Isn't. The Silent OT Security Crisis in Indian Manufacturing.

Picture this: A production line in Nagpur. Machines humming. Shifts are running on time. Orders going out.

Somewhere in that network, quietly, for the past 42 days, a ransomware group has been watching. Mapping your systems. Waiting for the right moment.

Then one morning, the machines stop.

This isn't fiction. This is the average dwell time for ransomware inside Operational Technology (OT) environments in 2025, 42 days before detection, according to Dragos' 2025 OT Cybersecurity Report. Forty-two days of invisible access. And most manufacturers never knew anyone was there.

### India Is Now APAC's Ransomware Epicentre

Let that sink in for a moment.

India has become the Asia-Pacific region's ransomware epicentre in 2025, with 65% of hit companies paying the ransom, making Indian manufacturers among the most lucrative targets on the planet. (Source: Industrial Cyber, April 2026)

The incidents aren't hypothetical:

-Raymond Limited, one of India's most recognised fabric manufacturers, faced an IT disruption caused by ransomware in early 2025, affecting operations.

Deck India Engineering Pvt. Ltd., a heat treatment firm, was targeted by Tengu ransomware, exposing sensitive operational data.



In 2020, a malware attack affecting Mumbai's power grid caused widespread outages, a stark demonstration of what OT compromise looks like at scale. (Source: Drishti IAS)

These aren't isolated incidents. They're a pattern and the pattern is accelerating.

## The Numbers Behind the Crisis

The scale of what's happening in industrial OT environments globally and specifically in India is staggering:

Ransomware groups targeting industrial organisations increased 64% year-on-year in 2025. Dragos tracked 119 active groups targeting OT, up from 80 in 2024, collectively hitting 3,300 industrial organisations. (Source: Security Brief / Dragos, February 2026)

- ICS vulnerability disclosures reached 2,451 in 2025, almost double the 1,690 seen in 2024, spanning 152 vendors. (Source: Cyble Research & Intelligence Labs Annual Threat Report, January 2026)

- Manufacturing accounted for more than two-thirds of all OT ransomware victims in 2025. (Source: Dragos)

- The industrial sector saw the sharpest increase in data breach costs of any sector in 2024, rising by \$830,000 per incident. (Source: Zero Networks / IBM)

And yet: only 19% of organisations felt completely prepared to handle OT security issues during the same period. (Source: Zero Networks, 2025)

## Why OT Security Is Fundamentally Different And Why IT Solutions Alone Won't Save You

Here's where many businesses get it wrong.



They assume their existing IT security stack — their firewall, their endpoint protection, their SIEM — covers their shop floor too. It doesn't.

OT environments have unique characteristics that make them a different beast entirely:

#### 1. Legacy Systems That Were Never Built for Connectivity

PLCs, SCADA systems, and industrial controllers that run your production line were often designed decades ago — long before anyone imagined they'd be connected to the internet. They can't run antivirus. They can't be patched without halting production. They are, by design, sitting ducks.

#### 2. The IT-OT Convergence Gap

India's rapid push toward Industry 4.0 and the Make in India initiative has connected factory floors to enterprise networks faster than security frameworks can adapt. The result: flat, unsegmented networks where a breach in IT becomes a breach in OT. \*(Source: EIMT, March 2026)\*

#### 3. Physical Consequences

Unlike a data breach that compromises files, an OT attack can physically stop machinery, damage equipment, disrupt supply chains, and — in critical infrastructure — risk human safety. The AIIMS Delhi ransomware incident didn't just lock down computers. It disrupted health services and affected medical supply chains for months.

#### 4. Compliance Is Now Mandatory, Not Optional

CERT-In's 2025 guidelines now mandate that OT/ICS environments are explicitly covered in audits. Cyber incidents



affecting OT must be reported within **\*\*6 hours\*\***. Annual audits are required. *\*(Source: Ogma, 2025)\**

The regulatory clock is ticking — and most manufacturers aren't ready.

### The Gap Is Glaring

60% of Indian manufacturers use IIoT devices. Only **\*\*25%\*\*** have robust OT security controls<sup>\*\*</sup>. *\*(Source: IDC 2024, via Whizhack 2025)\**

70% of Indian industrial firms lack adequate OT security altogether. *\*(Source: Gartner 2024)\**

The India OT security market is projected to reach **\*\*USD 1 billion by 2025\*\*** — yet investment continues to trail the threat curve. *\*(Source: Ogma, 2025)\**

This is a gap that attackers are actively and ruthlessly exploiting.

### What Proper IT-OT Security Looks Like in Practice

At CoreGenix, we work across both the IT and OT security landscape because threats don't respect the boundary between your server room and your shop floor. Here's what a comprehensive approach actually covers:

#### Network Segmentation & Micro-Segmentation

Your OT network should never share a flat network with corporate IT. Isolating PLCs, SCADA systems, and HMIs prevents lateral movement if IT is compromised.

#### Asset Visibility



You can't protect what you can't see. A complete inventory of every OT device from sensors to controllers is the starting point for any meaningful security posture.

### OT-Specific Threat Monitoring

Standard SIEM tools don't understand Modbus, OPC, DNP3, or IEC 104 — the industrial protocols your machines speak. OT threat detection requires purpose-built monitoring that understands operational context.

### Incident Response Planning for OT

In an IT breach, you take systems offline. In OT, taking systems offline can cost crores per hour. Your IR plan must account for the operational reality of your environment.

### VAPT for Industrial Environments

Vulnerability Assessment and Penetration Testing for OT is a discipline on its own probing industrial protocols, remote access points, and IT-OT integration layers for weaknesses before attackers find them.

### CERT-In Compliance & Audit Readiness

With mandatory 6-hour reporting windows and annual audits now required, your OT environment needs to be audit-ready at all times not just when an inspection is scheduled.

### The Make in India Opportunity, Secured or Squandered?

India's manufacturing economy is targeting \$1 trillion by 2030 (NITI Aayog). Smart factories, 5G-connected production, IIoT sensors, and Industry 4.0 automation are the engines of that ambition.



But every new connection is a new attack surface.

The businesses that will lead India's manufacturing renaissance won't just be the ones with the most advanced machines. They'll be the ones who protected those machines while everyone else assumed it couldn't happen to them.

The question isn't whether your OT environment is a target.

It already is. The question is whether you're protected when the attack arrives.

### CoreGenix: Bridging IT and OT Security

With 18+ years of experience in IT infrastructure and cybersecurity across Indian enterprises, CoreGenix offers end-to-end IT & OT security solutions — from initial assessments and network segmentation to continuous managed monitoring and CERT-In compliance support.

Because in 2025, the factory floor and the server room share the same threat landscape. It's time your security strategy did too.

Specializations: OT Security | VAPT | Managed SOC/NOC | IT Infrastructure | Cloud Security | InfoSec GRC & Audits | CERT-In Compliance

#OTSecurity #ICSecurity #ManufacturingCybersecurity  
#IndustrialCyber #Ransomware #MakeInIndia #Industry40  
#CriticalInfrastructure #CoreGenix #CybersecurityIndia #CERTIn  
#SCADASecurity #ITOTConvergence #VAPT\*\*

## Sources

1. Dragos / Security Brief — \*OT Cyber Threats Shift from Spying to Disruption in 2025\* (February 2026) — <https://securitybrief.co.uk/story/ot-cyber-threats-shift-from-spying-to-disruption-in-2025>
2. Industrial Cyber — \*Manufacturing Absorbs 56% Ransomware Surge of Global Attacks in 2025\* (April 2026) — <https://industrialcyber.co/manufacturing/manufacturing-absorbs-56-ransomware-surge-of-global-attacks-in-2025-as-raas-legacy-ot-supply-chains-fuel-spike/>
3. Cyble Research & Intelligence Labs — \*Annual Threat Landscape Report 2025\*, via Infosecurity Magazine (January 2026) — <https://www.infosecurity-magazine.com/news/cyber-threat-actors-ramp-up-ics/>
4. Zero Networks — \*OT Security Trends 2025: Escalating Threats and Evolving Tactics\* — <https://zeronetworks.com/blog/ot-security-trends-2025-escalating-threats-evolving-tactics>
5. EIMT — \*25 Major Cyber Attacks in India: Biggest Data Breaches and Lessons\* (March 2026) — <https://www.eimt.edu.eu/25-major-cyber-attacks-in-india-threats-and-strategies>
6. Ogma — \*OT/ICS Cyberattacks on Indian Manufacturing: 10 Real Incidents & How to Defend Your Shopfloor\* — <https://ogma.in/blog/ot-ics-cyberattacks-indian-manufacturing-defence-guide>
7. Whizhack — \*The Ultimate Guide to OT Security Products in India 2025\* — <https://whizhack.com/blog/the-ultimate-guide-to-ot-security-products-in-india-2025>
8. Drishti IAS — \*Strengthening India's Cyber Security Architecture\* — <https://www.drishtiias.com/daily-updates/daily-news-editorials/strengthening-indias-cyber-security-architecture>